

# **Software Requirements Specification (SRS)**

## **Online Exam Portal**

Project: Online Exam Portal

Team: Team 15

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## Revision history

| Version | Date       | Author  | Change summary                                                                                                                | Approval |
|---------|------------|---------|-------------------------------------------------------------------------------------------------------------------------------|----------|
|         |            |         |                                                                                                                               |          |
| 1.0     | 05-09-2026 | Team 15 | Merged supporting UML diagrams, system architecture and database design into a single self-contained SRS for final submission | Final    |

## Approvals

| Role               | Name                         | Signature / Email   | Date     |
|--------------------|------------------------------|---------------------|----------|
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|                    |                              |                     |          |

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# 1. Introduction

## 1.1 Purpose

This document is the complete, self-contained Software Requirements Specification (SRS) for the Online Exam Portal, a web-based system that lets an Admin (faculty/exam coordinator) build a reusable MCQ question bank, assemble and schedule exams from it, and lets Students attempt those exams under a server-enforced timer with automatic evaluation and reporting. It defines functional and non-functional requirements, interfaces, security requirements, system models/diagrams, database design, and verification criteria for the development team and course evaluators. No separate report or appendix is required to understand the system.

## 1.2 Scope

Covers Admin and Student web interactions: authentication, question-bank management, exam creation/scheduling, timed exam attempts with auto-save, server-side auto-evaluation, result/analytics reporting, bulk student import, and basic tab-switch proctoring signals. Excludes descriptive/subjective-question grading, live webcam proctoring, plagiarism detection, and integration with external university ERP/payment systems.

## 1.3 Audience

Developers, QA testers (Team 15), course instructor/evaluators, and prospective student/admin users participating in a pilot/demo.

## 1.4 Definitions

MCQ (Multiple Choice Question), RBAC (Role-Based Access Control), JWT (JSON Web Token), NFR (Non-Functional Requirement), FR (Functional Requirement), RTM (Requirements Traceability Matrix), TLS (Transport Layer Security), CSV (Comma-Separated Values), WCAG (Web Content Accessibility Guidelines), UML (Unified Modeling Language), ERD (Entity-Relationship Diagram), PK/FK (Primary Key/Foreign Key).

# 2. Overall description

## 2.1 Product perspective

The Online Exam Portal is a standalone, self-hosted three-tier web application (client, application server, database) used within a single department/institution. It is not a component of a larger commercial LMS; it interacts only with an SMTP service for email notifications. The full system architecture is presented as Figure 6 in Section 7.6.

## 2.2 Major product functions (detailed)

- Admin/Student authentication with RBAC
- Question bank management (create/edit/delete, tagging)
- Exam creation, question selection (manual/random) and scheduling
- Batch/section-based exam assignment and student bulk import
- Timed exam attempt with auto-save and server-enforced auto-submit
- Automatic server-side evaluation with negative marking
- Result and class/question-wise analytics reporting (CSV/PDF export)
- Tab-switch/focus-loss proctoring-lite logging and admin review
- Email notifications for exam scheduling and result declaration

## 2.3 User roles and characteristics (expanded)

- Admin (Faculty/Exam Coordinator): comfortable with spreadsheets/CSV; needs an efficient question-bank workflow and clear analytics; trusted role, not adversarial.

- Student: basic computer literacy; expects a clear timer, no ambiguity about time remaining, and instant, fair results.

## 2.4 Operating environment

Modern evergreen browsers (Chrome, Firefox, Edge) on desktop and mobile; server deployed on a Linux VM/container (Docker) with PostgreSQL; institutional or campus Wi-Fi/LAN network.

## 2.5 Constraints

Must be deliverable within one academic semester by a 3-person student team; modest infrastructure budget (single VM or free-tier cloud instance for the demo); browser-based proctoring cannot detect a second physical device — stated as an explicit, accepted limitation.

# 3. External interface requirements

## 3.1 User interfaces

Primary UI: responsive web pages (desktop/mobile browser) with a persistent, always-visible countdown timer during an attempt; high-contrast mode and keyboard navigation for accessibility.

## 3.2 Hardware interfaces

None beyond a standard client device (desktop/laptop/tablet) with keyboard/mouse or touchscreen and a working network interface. No specialized hardware is required.

## 3.3 Software interfaces

- Application API (Django REST Framework or Express.js) over HTTPS/JSON for all client-server communication
- SMTP service (e.g., SendGrid/institutional mail server) for notification emails
- PostgreSQL (or MongoDB, MERN option) as the system of record
- Optional Redis cache for exam-session state and rate limiting

## 3.4 Communications

TLS 1.2+ enforced on all connections; auto-save requests use short-timeout retries with exponential backoff so a brief network blip does not fail the attempt.

# 4. System features (detailed)

*Each requirement below includes acceptance criteria and a reference test case. IDs follow OEP-F-####.*

## 4.1 Authentication & Access Control

*Description: Secure registration/login for Admin and Student roles with role-based access control (RBAC) enforced server-side.*

| Req ID    | Requirement (shall...)                                                                                                   | Type       | Priorty | Source/Stakeholder | Acceptance criteria / Test case ref                                                         | Comments / Dependencies           |
|-----------|--------------------------------------------------------------------------------------------------------------------------|------------|---------|--------------------|---------------------------------------------------------------------------------------------|-----------------------------------|
| OEP-F-001 | The system shall allow Admin and Student users to log in with email + password, issuing a role-scoped session/JWT token. | Functional | High    | All admin/student  | AC-OEP-F-001: Valid credentials return a token scoped to the correct role. Test: TC-Auth-01 | Passwords hashed (see OEP-SR-001) |
| OEP-F-002 | The system shall lock a login after 5 consecutive failed attempts for 15 minutes and log the event.                      | Functional | High    | Security           | AC-OEP-F-002: 6th attempt is rejected regardless of correctness. Test: TC-Auth-02           |                                   |

| Req ID    | Requirement (shall...)                                                                               | Type       | Priority | Source/Stakeholder | Acceptance criteria / Test case ref                                                  | Comments / Dependencies |
|-----------|------------------------------------------------------------------------------------------------------|------------|----------|--------------------|--------------------------------------------------------------------------------------|-------------------------|
| OEP-F-003 | The system shall enforce RBAC at the API level so Student tokens cannot access Admin-only endpoints. | Functional | High     | Security           | AC-OEP-F-003: Student token calling an admin route returns 403. Test: TC-Sec-Auth-01 | See OEP-SR-004          |
| OEP-F-004 | The system shall allow an Admin to reset a student's password and force re-verification via email.   | Functional | Medium   | Admin              | AC-OEP-F-004: Reset link expires after 1 hour and works once. Test: TC-Auth-03       |                         |

## 4.2 Question Bank Management

*Description: Admin builds and maintains a reusable bank of MCQ questions independent of any single exam.*

| Req ID    | Requirement (shall...)                                                                                                                                                       | Type       | Priority | Source/Stakeholder | Acceptance criteria / Test case ref                                                          | Comments / Dependencies |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|----------|--------------------|----------------------------------------------------------------------------------------------|-------------------------|
| OEP-F-005 | The system shall allow an Admin to create/edit/delete MCQ questions with 2-6 options, exactly one marked correct, marks, negative marks, subject, topic and difficulty tags. | Functional | High     | Admin/Faculty      | AC-OEP-F-005: A question with zero or 2+ correct options is rejected on save. Test: TC-QB-01 |                         |
| OEP-F-006 | The system shall allow an Admin to search/filter questions by subject, topic and difficulty.                                                                                 | Functional | Medium   | Admin/Faculty      | AC-OEP-F-006: Filter returns only matching questions. Test: TC-QB-02                         |                         |
| OEP-F-007 | The system shall never expose a question's correct-option flag to any Student-facing API response.                                                                           | Functional | High     | Security           | AC-OEP-F-007: /attempts/{id}/questions response contains no isCorrect field. Test: TC-Sec-01 | See OEP-SR-003          |

## 4.3 Exam Creation & Scheduling

*Description: Admin assembles an exam from the question bank and schedules a fixed attempt window for one or more student batches.*

| Req ID    | Requirement (shall...)                                                                                                                  | Type       | Priority | Source/Stakeholder | Acceptance criteria / Test case ref                                                          | Comments / Dependencies |
|-----------|-----------------------------------------------------------------------------------------------------------------------------------------|------------|----------|--------------------|----------------------------------------------------------------------------------------------|-------------------------|
| OEP-F-008 | The system shall allow an Admin to create an exam with title, duration, total marks, start time, end time and result-visible-from time. | Functional | High     | Admin/Faculty      | AC-OEP-F-008: Exam saved with all fields validated (end > start). Test: TC-Exam-01           |                         |
| OEP-F-009 | The system shall allow an Admin to attach questions to an exam manually or via random selection from tagged pools.                      | Functional | High     | Admin/Faculty      | AC-OEP-F-009: Random selection respects requested count and difficulty mix. Test: TC-Exam-02 | Depends on OEP-F-005    |
| OEP-F-010 | The system shall allow an Admin to assign a scheduled exam to one or more student batches.                                              | Functional | High     | Admin/Faculty      | AC-OEP-F-010: Only students in an assigned batch can see the exam. Test: TC-Exam-03          |                         |

## 4.4 Student & Batch Management

*Description: Admin maintains the student roster and batch/section groupings used for exam assignment.*

| Req ID    | Requirement (shall...)                                                                                  | Type       | Priorty | Source/Stake holder | Acceptance criteria / Test case ref                                                                                     | Comments / Dependenci es |
|-----------|---------------------------------------------------------------------------------------------------------|------------|---------|---------------------|-------------------------------------------------------------------------------------------------------------------------|--------------------------|
| OEP-F-011 | The system shall allow an Admin to bulk-import students via a CSV/Excel file (name, email, USN, batch). | Functional | Medium  | Admin               | AC-OEP-F-011: Valid rows create accounts; invalid rows are reported with reasons, not silently dropped. Test: TC-Stu-01 |                          |
| OEP-F-012 | The system shall allow an Admin to create batches/sections and map students to them (many-to-many).     | Functional | Medium  | Admin               | AC-OEP-F-012: A student can belong to more than one batch. Test: TC-Stu-02                                              |                          |

## 4.5 Exam Attempt Engine

*Description: Enforces a fair, server-timed exam experience for the student, resilient to refresh/crash.*

| Req ID    | Requirement (shall...)                                                                                                | Type       | Priorty | Source/Stake holder         | Acceptance criteria / Test case ref                                                                                  | Comments / Dependenci es |
|-----------|-----------------------------------------------------------------------------------------------------------------------|------------|---------|-----------------------------|----------------------------------------------------------------------------------------------------------------------|--------------------------|
| OEP-F-013 | The system shall show a Student only the exams assigned to their batch(es) and currently within the scheduled window. | Functional | High    | Student                     | AC-OEP-F-013: An exam outside the window is not listed/accessible. Test: TC-Attempt-00                               |                          |
| OEP-F-014 | The system shall allow only one active attempt per (exam, student) pair, blocking a second concurrent session.        | Functional | High    | Security/Academic Integrity | AC-OEP-F-014: Second login for the same exam is rejected while the first attempt is IN_PROGRESS. Test: TC-Attempt-04 | See OEP-SR-005           |
| OEP-F-015 | The system shall auto-save the student's current answer state to the server every 10 seconds during an attempt.       | Functional | High    | Student                     | AC-OEP-F-015: Refreshing mid-exam restores the last auto-saved answers. Test: TC-Attempt-01                          |                          |
| OEP-F-016 | The system shall auto-submit an attempt the instant the server-side timer reaches zero, independent of client clock.  | Functional | High    | Academic Integrity          | AC-OEP-F-016: Attempt status becomes AUTO_SUBMITTED at expiry even if the tab is inactive. Test: TC-Attempt-02       |                          |

## 4.6 Evaluation & Results

*Description: Automatic, tamper-resistant scoring performed entirely server-side upon submission.*

| Req ID    | Requirement (shall...)                                                                                                                                | Type       | Priorty | Source/Stake holder | Acceptance criteria / Test case ref                                                                        | Comments / Dependenci es |
|-----------|-------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------|---------------------|------------------------------------------------------------------------------------------------------------|--------------------------|
| OEP-F-017 | The system shall auto-evaluate a submitted attempt by comparing each Response to the stored correct Option and applying marks/negative-marking rules. | Functional | High    | Academic/Business   | AC-OEP-F-017: Score = sum(correct marks) - sum(incorrect negative marks); unanswered = 0. Test: TC-Eval-01 |                          |

| Req ID    | Requirement (shall...)                                                                                          | Type       | Priority | Source/Stakeholder | Acceptance criteria / Test case ref                                                           | Comments / Dependencies |
|-----------|-----------------------------------------------------------------------------------------------------------------|------------|----------|--------------------|-----------------------------------------------------------------------------------------------|-------------------------|
| OEP-F-018 | The system shall lock all Responses of an attempt from further modification once its status leaves IN PROGRESS. | Functional | High     | Security           | AC-OEP-F-018: A write attempt to a submitted attempt's responses is rejected. Test: TC-Sec-02 |                         |

## 4.7 Reporting, Proctoring-lite & Notifications

*Description: Analytics, integrity signals and notifications built on top of the core attempt/result data.*

| Req ID    | Requirement (shall...)                                                                                                        | Type       | Priority | Source/Stakeholder | Acceptance criteria / Test case ref                                                                               | Comments / Dependencies |
|-----------|-------------------------------------------------------------------------------------------------------------------------------|------------|----------|--------------------|-------------------------------------------------------------------------------------------------------------------|-------------------------|
| OEP-F-019 | The system shall generate, per exam, a class performance report and a question-wise difficulty report, exportable to CSV/PDF. | Functional | Medium   | Admin/Faculty      | AC-OEP-F-019: Export file matches on-screen report totals. Test: TC-Rep-01                                        |                         |
| OEP-F-020 | The system shall log tab-switch/window-blur events per attempt and present a flagged-attempts view to the Admin.              | Functional | Medium   | Academic Integrity | AC-OEP-F-020: Two tab switches during an attempt create two ProctoringLog rows visible to Admin. Test: TC-Proc-01 |                         |
| OEP-F-021 | The system shall allow an Admin to manually disqualify or re-permit a flagged attempt, recording the action in an audit log.  | Functional | Medium   | Admin              | AC-OEP-F-021: A disqualified attempt is excluded from the result/report until re-permitted. Test: TC-Proc-02      | See OEP-SR-006          |
| OEP-F-022 | The system shall send an email notification to affected students when an exam is scheduled and when results become visible.   | Functional | Low      | Student            | AC-OEP-F-022: Notification email is queued within 1 minute of the triggering action. Test: TC-Notif-01            | Async job               |

## 5. Non-functional requirements (detailed)

*NFRs below are measurable and tied to test plans. IDs OEP-NF-####.*

| Req ID     | Requirement                                                                                                                                                                   | Category    | Priority | Acceptance criteria / Measurement                                           |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|-----------------------------------------------------------------------------|
| OEP-NF-001 | The exam attempt page shall load within 2 seconds and support at least 200 concurrent students per exam without response-time degradation beyond 3 seconds (95th percentile). | Performance | High     | Load test with 200 simulated students. Test: TC-Perf-01                     |
| OEP-NF-002 | The application tier shall be horizontally scalable and the database indexed on (exam_id, student_id) for attempts and responses.                                             | Scalability | Medium   | Design/code review + query plan (EXPLAIN) shows index usage.                |
| OEP-NF-003 | The system shall provide 99.5% availability during any published exam window, excluding pre-announced maintenance.                                                            | Reliability | High     | Uptime monitoring logs during test exam windows.                            |
| OEP-NF-004 | The system shall auto-save attempt state at most 10 seconds apart so that a client                                                                                            | Reliability | High     | Simulate crash mid-attempt; verify recovered state age. Test: TC-Attempt-01 |



| Req ID     | Requirement                                                                                                                                                       | Category                | Priority | Acceptance criteria / Measurement                                        |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|----------|--------------------------------------------------------------------------|
|            | crash loses no more than the last 10 seconds of activity.                                                                                                         |                         |          |                                                                          |
| OEP-NF-005 | The UI shall be responsive across desktop and mobile-browser viewports and shall meet WCAG 2.1 AA basics (contrast, keyboard navigation) for exam-taking screens. | Usability/Accessibility | Medium   | Manual accessibility audit + Lighthouse score $\geq 90$ . Test: TC-UX-01 |
| OEP-NF-006 | Core modules (evaluation engine, exam engine) shall carry $\geq 70\%$ automated unit-test coverage.                                                               | Maintainability         | Medium   | Coverage report from CI pipeline.                                        |

## 5.1 Security

### 5.1.1 Security Objectives

1. Ensure the exam question bank and answer keys remain confidential and are never exposed to a Student via any client-accessible channel (API response, browser dev tools, or page source), before or during an exam.
2. Guarantee the integrity and non-repudiation of submitted answers and computed results: once an attempt leaves IN\_PROGRESS status, no student and no unauthorized party can alter its Responses, and every Admin override is attributable via an audit log.
3. Protect user credentials and personally identifiable information (name, email, USN) against unauthorized access through hashing, encryption in transit, and role-based access control.

### 5.1.2 Security Requirements

| Req ID     | Requirement (shall...)                                                                                                                                                            | Type     | Priority | Acceptance criteria / Test case ref                                                                       |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|----------|-----------------------------------------------------------------------------------------------------------|
| OEP-SR-001 | The system shall store all passwords using a salted one-way hash (bcrypt or Argon2); plaintext passwords shall never be stored or logged.                                         | Security | High     | Inspect DB directly; confirm no plaintext. Test: TC-Sec-Pwd-01                                            |
| OEP-SR-002 | All client-server traffic shall be served over HTTPS/TLS 1.2+, with session cookies marked HttpOnly and Secure.                                                                   | Security | High     | TLS scan + cookie attribute inspection.                                                                   |
| OEP-SR-003 | The correct-option flag and answer key shall never be included in any API response reachable by a Student role; evaluation shall occur only in the server-side Evaluation Engine. | Security | High     | Inspect network responses during an attempt. Test: TC-Sec-01                                              |
| OEP-SR-004 | Role-based access control shall be enforced on every API endpoint server-side (not only hidden in the UI).                                                                        | Security | High     | Attempt admin endpoints with a student token; expect 403. Test: TC-Sec-Auth-01                            |
| OEP-SR-005 | Each exam attempt shall be bound to a single active session; a second login for the same (exam, student) pair while IN_PROGRESS shall be rejected.                                | Security | High     | Concurrent login test. Test: TC-Attempt-04                                                                |
| OEP-SR-006 | All Admin actions that alter question content, results, or attempt status (disqualify/re-permit) shall be written to an append-only audit log with actor, timestamp and action.   | Security | Medium   | Perform an override; verify immutable log entry. Test: TC-Sec-Audit-01                                    |
| OEP-SR-007 | All user-supplied input shall be validated and persisted via parameterized queries/ORM to prevent SQL/NoSQL injection and stored XSS in rendered question text.                   | Security | High     | Run OWASP ZAP / manual injection payloads against search and question-text fields. Test: TC-Sec-Inject-01 |

## 6. Quality attributes & Acceptance tests

- Exit criteria for acceptance: all High-priority functional requirements implemented and verified, no critical NFR or security-requirement failures, and the RTM (Section 9) shows all associated test cases passed.

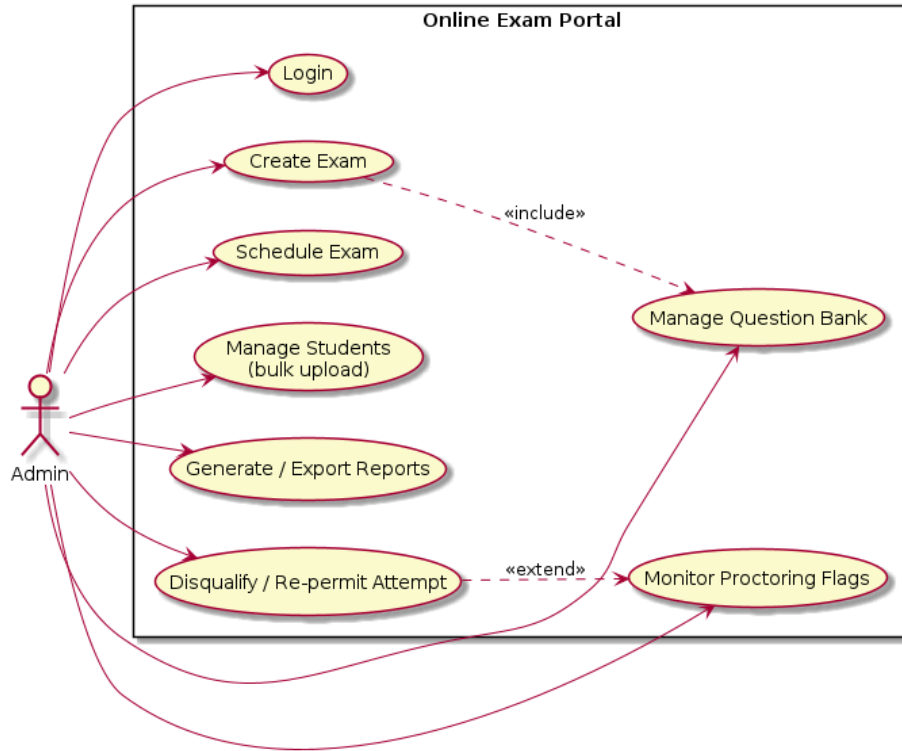
- Acceptance test suites: Authentication & RBAC, Question Bank, Exam Scheduling, Exam Attempt (timer/auto-save/auto-submit), Evaluation, Reporting, Proctoring-lite, Performance, and Security.

## 7. System Models and Diagrams

This section presents the complete set of UML models and the system architecture diagram for the Online Exam Portal. Each diagram is numbered, titled, and followed by a short explanation of its relevance to the system so that this SRS is sufficient on its own to understand the system's structure and behaviour.

### 7.1 Use Case Diagram — Admin

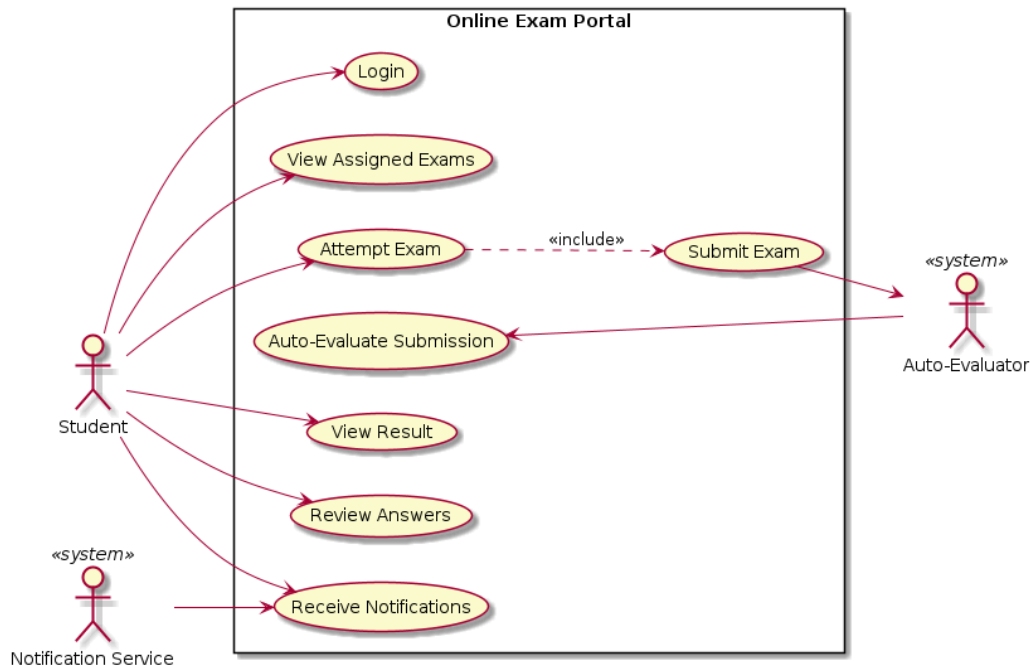
**Figure 1: Admin Use Case Diagram**



*Shows the Admin actor's interactions with the system: logging in, managing the question bank, creating and scheduling exams, managing students, generating reports, and reviewing/acting on proctoring flags. The <<include>> relationship shows that creating an exam always draws on the question bank, and the <<extend>> relationship shows that disqualifying an attempt is an optional extension of monitoring proctoring flags, not a mandatory step.*

### 7.2 Use Case Diagram — Student

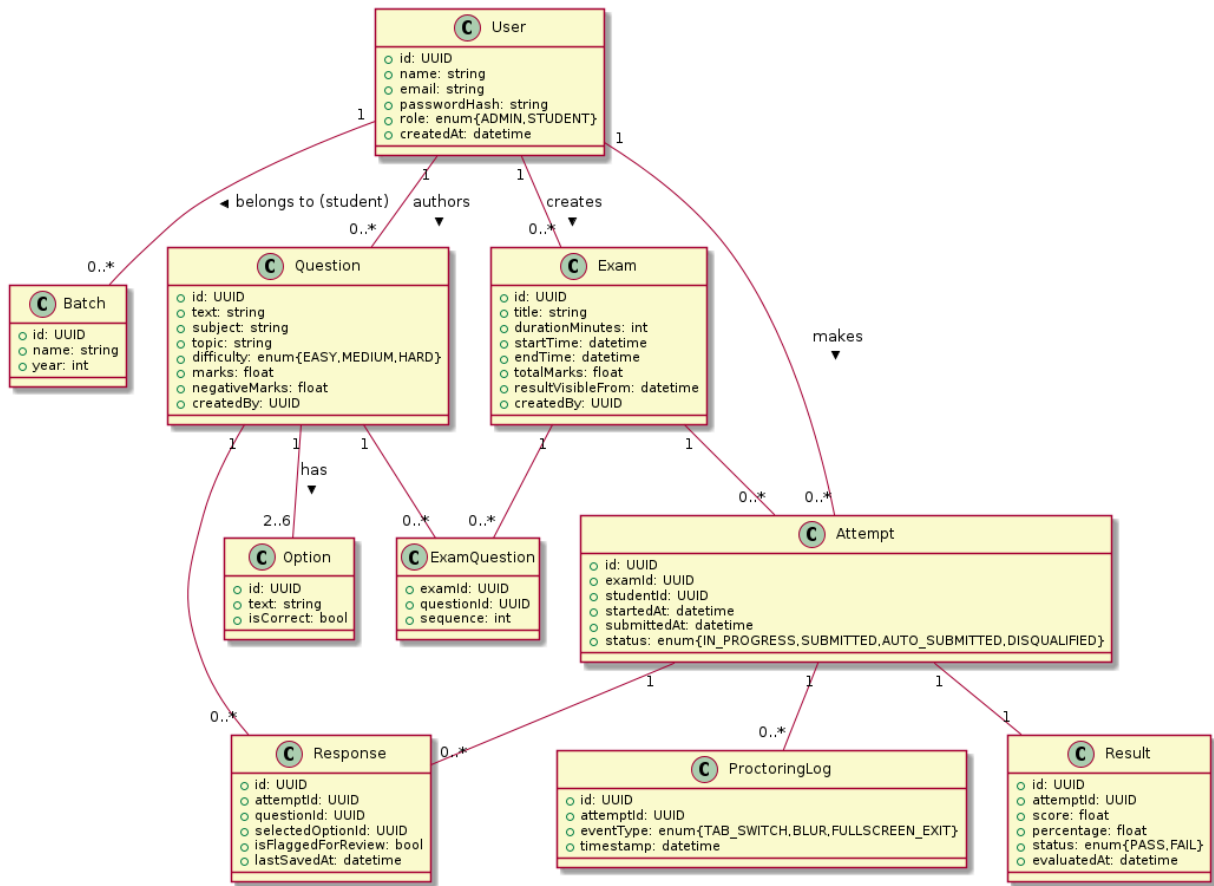
**Figure 2: Student Use Case Diagram**



*Shows the Student actor's journey: logging in, viewing assigned exams, attempting an exam (which includes submitting it), and viewing/reviewing results. Two system actors are modeled explicitly — the Auto-Evaluator, which scores a submission the moment it is submitted, and the Notification Service, which delivers schedule and result alerts — to make clear that evaluation and notification are automated, not manual admin steps.*

## 7.3 Class Diagram

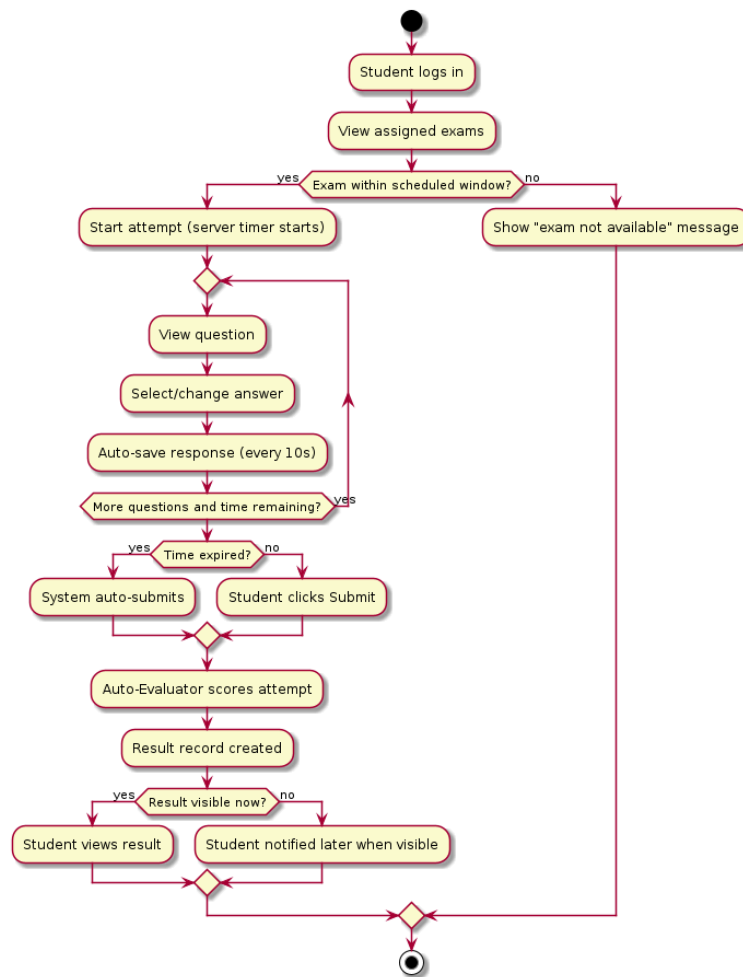
**Figure 3: Domain Class Diagram**



*Models the core domain entities and their relationships: a User (Admin or Student) authors Questions and Exams and makes Attempts; an Exam is composed of Questions through the ExamQuestion association; each Question owns 2–6 Options with exactly one correct; each Attempt owns many Responses (one per question), exactly one Result, and zero or more ProctoringLog entries. This diagram is the direct blueprint for the relational schema in Section 8.*

## 7.4 Activity Diagram

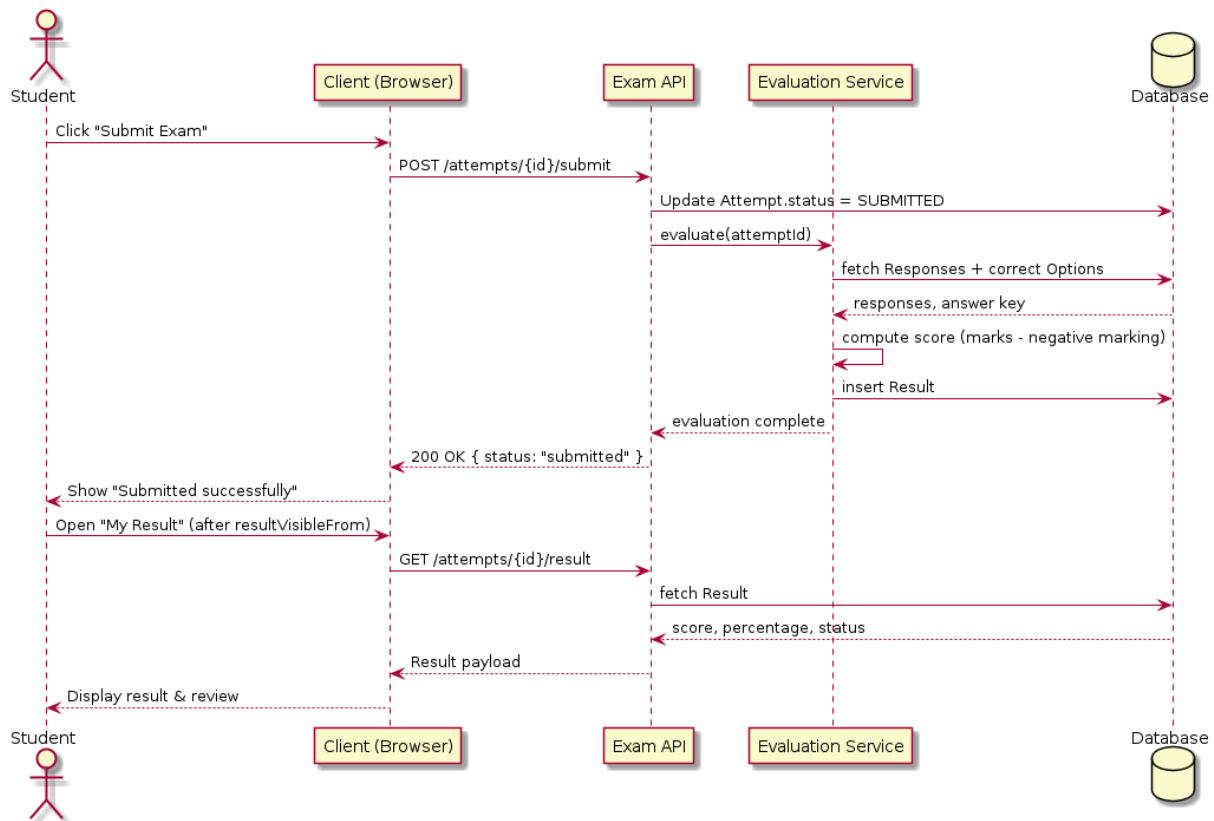
**Figure 4: Student Exam-Attempt Activity Diagram**



Traces the control flow a Student follows from login to result: the system first checks whether the exam is within its scheduled window; if so, the student answers questions with continuous auto-save until either the timer expires (triggering an automatic submission) or the student submits manually; the Auto-Evaluator then scores the attempt, after which the result is shown immediately or deferred until the admin-configured result-visible time.

## 7.5 Sequence Diagram

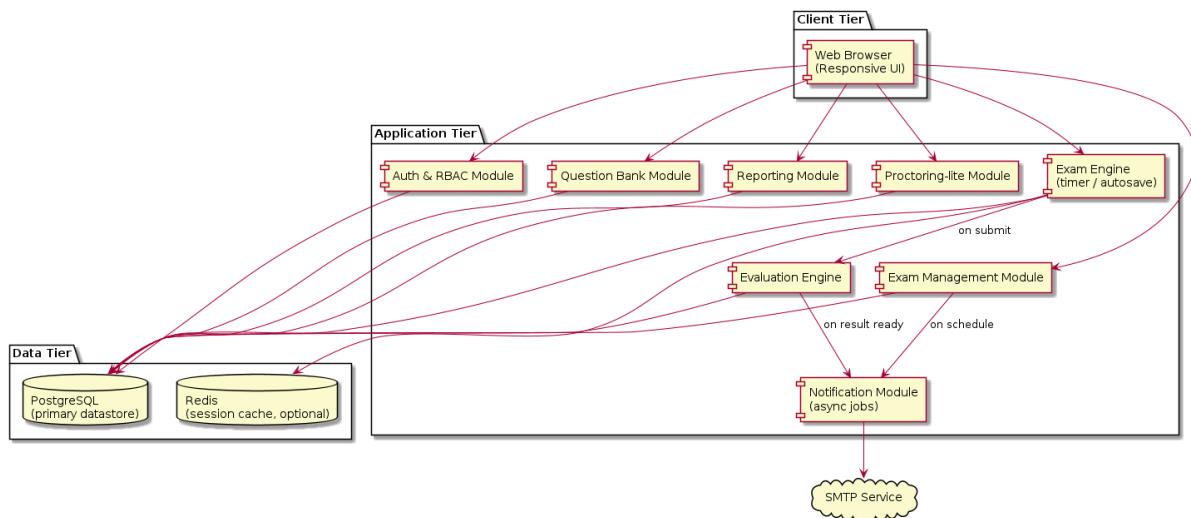
**Figure 5: Submit & Auto-Evaluate Sequence Diagram**



Details the message flow between the Student, the browser client, the Exam API, the Evaluation Service and the database at the most security-critical moment in the system: submission and scoring. It shows that the answer key is fetched and compared only inside the server-side Evaluation Service — never sent to the client — directly implementing security requirement OEP-SR-003.

## 7.6 System Architecture Diagram

Figure 6: Three-Tier System Architecture



Depicts the Client Tier (responsive browser UI), Application Tier (Auth & RBAC, Question Bank, Exam Management, Exam Engine, Evaluation Engine, Reporting, Proctoring-lite and Notification modules) and Data Tier (PostgreSQL as the system of record, with an optional Redis cache for session/timer state), plus the external SMTP dependency for notifications. This is the physical/logical realization of the module breakdown referenced throughout Sections 4 and 5.





## 8. Database Design

The relational schema below directly realizes the Class Diagram in Section 7.3. All tables are normalized to Third Normal Form (3NF), with one documented, deliberate denormalization for read-performance (Section 8.3).

### 8.1 Entities Overview

Users, Batches, StudentBatches (join), Questions, Options, Exams, ExamQuestions (join), ExamBatches (join), Attempts, Responses, Results, ProctoringLogs. A User with role STUDENT belongs to one or more Batches via StudentBatches; a User with role ADMIN authors many Questions and Exams; an Exam is composed of many Questions through ExamQuestions and is assigned to one or more Batches through ExamBatches; a Student makes one Attempt per Exam (unique on exam\_id + student\_id), each with many Responses and exactly one Result.

### 8.2 Table Definitions (Keys & Attributes)

#### users

| Column        | Type                    | Key               |
|---------------|-------------------------|-------------------|
| id            | UUID                    | PK                |
| name          | VARCHAR(100)            |                   |
| email         | VARCHAR(150) UNIQUE     |                   |
| password_hash | VARCHAR(255)            |                   |
| role          | ENUM('ADMIN','STUDENT') |                   |
| usn           | VARCHAR(20) NULL        | UNIQUE (students) |
| created_at    | TIMESTAMP               |                   |

#### batches

| Column | Type        | Key |
|--------|-------------|-----|
| id     | UUID        | PK  |
| name   | VARCHAR(50) |     |
| year   | INT         |     |

#### student\_batches

| Column     | Type | Key                 |
|------------|------|---------------------|
| student_id | UUID | PK, FK → users.id   |
| batch_id   | UUID | PK, FK → batches.id |

#### questions

| Column         | Type                         | Key           |
|----------------|------------------------------|---------------|
| id             | UUID                         | PK            |
| text           | TEXT                         |               |
| subject        | VARCHAR(50)                  |               |
| topic          | VARCHAR(50)                  |               |
| difficulty     | ENUM('EASY','MEDIUM','HARD') |               |
| default_marks  | DECIMAL(5,2)                 |               |
| negative_marks | DECIMAL(5,2)                 |               |
| created_by     | UUID                         | FK → users.id |
| created_at     | TIMESTAMP                    |               |

#### options

| Column      | Type    | Key               |
|-------------|---------|-------------------|
| id          | UUID    | PK                |
| question_id | UUID    | FK → questions.id |
| text        | TEXT    |                   |
| is_correct  | BOOLEAN |                   |

### exams

| Column              | Type         | Key           |
|---------------------|--------------|---------------|
| id                  | UUID         | PK            |
| title               | VARCHAR(150) |               |
| duration_minutes    | INT          |               |
| start_time          | TIMESTAMP    |               |
| end_time            | TIMESTAMP    |               |
| total_marks         | DECIMAL(6,2) |               |
| result_visible_from | TIMESTAMP    |               |
| created_by          | UUID         | FK → users.id |

### exam\_questions

| Column      | Type | Key                   |
|-------------|------|-----------------------|
| exam_id     | UUID | PK, FK → exams.id     |
| question_id | UUID | PK, FK → questions.id |
| sequence    | INT  |                       |

### exam\_batches

| Column   | Type | Key                 |
|----------|------|---------------------|
| exam_id  | UUID | PK, FK → exams.id   |
| batch_id | UUID | PK, FK → batches.id |

### attempts

| Column       | Type                                                            | Key                              |
|--------------|-----------------------------------------------------------------|----------------------------------|
| id           | UUID                                                            | PK                               |
| exam_id      | UUID                                                            | FK → exams.id                    |
| student_id   | UUID                                                            | FK → users.id                    |
| started_at   | TIMESTAMP                                                       |                                  |
| submitted_at | TIMESTAMP NULL                                                  |                                  |
| status       | ENUM('IN_PROGRESS','SUBMITTED','AUTO_SUBMITTED','DISQUALIFIED') |                                  |
| (unique)     | UNIQUE(exam_id, student_id)                                     | one attempt per student per exam |

### responses

| Column             | Type      | Key               |
|--------------------|-----------|-------------------|
| id                 | UUID      | PK                |
| attempt_id         | UUID      | FK → attempts.id  |
| question_id        | UUID      | FK → questions.id |
| selected_option_id | UUID NULL | FK → options.id   |
| is_flagged         | BOOLEAN   |                   |
| last_saved_at      | TIMESTAMP |                   |

### results

| Column       | Type                | Key              |
|--------------|---------------------|------------------|
| id           | UUID                | PK               |
| attempt_id   | UUID UNIQUE         | FK → attempts.id |
| score        | DECIMAL(6,2)        |                  |
| percentage   | DECIMAL(5,2)        |                  |
| status       | ENUM('PASS','FAIL') |                  |
| evaluated_at | TIMESTAMP           |                  |

### proctoring\_logs

| Column     | Type                                         | Key              |
|------------|----------------------------------------------|------------------|
| id         | UUID                                         | PK               |
| attempt_id | UUID                                         | FK → attempts.id |
| event_type | ENUM('TAB_SWITCH','BLUR','FULL_SCREEN_EXIT') |                  |
| timestamp  | TIMESTAMP                                    |                  |

## 8.3 Normalization Notes

The schema is in Third Normal Form (3NF): every non-key attribute depends only on its own table's primary key, join tables (student\_batches, exam\_questions, exam\_batches) carry composite keys of exactly the two foreign keys they join with no other dependent data, and there are no transitive dependencies. The one deliberate exception is exams.total\_marks, which is a stored, denormalized summary recomputed whenever exam\_questions changes — a documented performance trade-off, since total\_marks is read on every result view but changes only when an admin edits the question list.

## 9. Requirements Traceability Matrix (RTM)

Every requirement ID below traces to the section defining it, the implementing module, and its verifying test case(s), confirming full traceability from requirement through design to test.

| Req ID     | Requirement (short)                    | Section ref / Design Spec | Module                          | Test case(s)               | Status (N/P/A) |
|------------|----------------------------------------|---------------------------|---------------------------------|----------------------------|----------------|
| OEP-F-001  | Login with RBAC                        | 4.1                       | AuthModule                      | TC-Auth-01, TC-Sec-Auth-01 | N              |
| OEP-F-005  | Question bank CRUD                     | 4.2                       | QuestionBankModule              | TC-QB-01                   | N              |
| OEP-F-007  | Hide answer key from student API       | 4.2 / OEP-SR-003          | QuestionBankModule / ExamEngine | TC-Sec-01                  | N              |
| OEP-F-009  | Build exam from question bank          | 4.3                       | ExamManagementModule            | TC-Exam-02                 | N              |
| OEP-F-014  | Single active attempt per exam         | 4.5 / OEP-SR-005          | ExamEngine                      | TC-Attempt-04              | N              |
| OEP-F-015  | Auto-save every 10s                    | 4.5                       | ExamEngine                      | TC-Attempt-01              | N              |
| OEP-F-016  | Server-side auto-submit on timeout     | 4.5                       | ExamEngine                      | TC-Attempt-02              | N              |
| OEP-F-017  | Auto-evaluate with negative marking    | 4.6                       | EvaluationEngine                | TC-Eval-01                 | N              |
| OEP-F-019  | Class & question-wise reports          | 4.7                       | ReportingModule                 | TC-Rep-01                  | N              |
| OEP-F-020  | Tab-switch proctoring log              | 4.7                       | ProctoringModule                | TC-Proc-01                 | N              |
| OEP-NF-001 | Response time / concurrency target     | 5                         | ExamEngine / Infra              | TC-Perf-01                 | N              |
| OEP-NF-003 | 99.5% availability during exam windows | 5                         | Infra / Ops                     | Ops monitoring             | N              |
| OEP-SR-001 | Password hashing                       | 5.1.2                     | AuthModule                      | TC-Sec-Pwd-01              | N              |
| OEP-SR-003 | Answer key never sent to client        | 5.1.2                     | ExamEngine                      | TC-Sec-01                  | N              |
| OEP-SR-005 | Single-session-per-attempt enforcement | 5.1.2                     | ExamEngine                      | TC-Attempt-04              | N              |